

Developing biomass energy from agricultural by-products in the context of trade development[☆]

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ABSTRACT

With global trade development, emerging economies can absorb biomass energy production techniques and technologies from developed countries. However, several constraints still need to be improved in producing biomass energy in emerging economies. Therefore, this research aims to understand the critical motivating factors of biomass energy production that contribute to a circular economy via agricultural by-products in Vietnam's emerging economy. This study employs a qualitative analysis using information from the in-depth interviews of farmer households, agricultural by-product traders, and bioenergy producers in Vietnam. This study revealed the motivating factors as the main determinants in producing biomass energy: (1) digitalization, (2) technical and logistics, (3) social capital, (4) prices, and (5) government supports. This study also confirms that Vietnam has diverse biomass waste sources and a high potential for biomass energy production. To motivate biomass energy development, the government should support credit and loans for stakeholders in producing biomass energy. Moreover, the government should encourage and support energy producers to implement digital transformation to expand production scale and improve business operations. Insights from this study can help farmers, traders, and bioenergy producers by adapting their strategies within their local contexts.

1. Introduction

Global energy consumption has increased rapidly in recent decades due to the development of the economy and people's needs [1]. To meet this energy demand, energy sources are exploited, such as fossil fuels, nuclear energy, and renewable energy, which fossil fuels account for most present in global energy [2]. Using fossil fuel sources such as coal, oil, and natural gas contributes the most to the increase in global greenhouse gas emissions and is the leading cause of climate change and global warming [3]. Therefore, using renewable energy to meet rising energy demand and mitigate climate change is one of the current global issues [4]. Among the forms of renewable energy, biomass energy production is one of the most potential energy sources due to its carbon neutrality, large output, environmental friendliness, and ease of development [5]. Consequently, this paper explores the motivating factors for developing biomass energy from agricultural by-products in an emerging economy.

Previous scholars have studied renewable energy and biomass

energy production from different materials and methods [5–7]. For example, Miao and Khanna [8] investigated the effectiveness of biomass programs under behavioral factors, credit constraints, and program design. He et al. [9] examined the environmental influence of biomass power compared to wind and solar power. Some scholars have analyzed biomass energy production from agricultural by-products [10,11]. They revealed the possibility of using agricultural by-products as biomass energy production, the sustainable, renewable use of natural resources, and turning waste into resources. These mentioned studies examined the technical approach to analyzing biomass production with the involvement of stakeholders from an economic perspective. Therefore, there is limited research analyzing driving factors influencing biomass production.

In term of macro level, contemporary research has raised the call for developing the bioeconomy which relies on renewable biological resources [12–16]. To shape the renewable-based economy, the role of institutions become importance in emphasizing the political stability, the control of corruption, rule of law, government effectiveness, voice

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and accountability [12]. In high-income countries, the institutional quality plays a critical role in improving the countries' environmental efficiency [16]. Meanwhile, developing countries have facing the challenges in developing bioeconomy because of fiscal restrictions, institutional barriers, and technology limitations [14]. Therefore, studying the biomass energy development in developing countries is essential to reduce the negative consequences of fossil fuel consumption and develop the bioeconomy.

Vietnam has the potential to develop biomass energy, especially from agricultural by-products. Wood fuel in Vietnam is about 75–80 million tons/year, equivalent to 26–28 million tons of oil/year (Hu; ng, 2017). The volume of biomass from agricultural by-products such as rice straw, rice husk, bagasse, and other wastes is about 30 million tons/year, equivalent to 10 million tons of oil/year (Vi; nh, 2020). The primary source of biomass is wood and crop by-products, including natural forests, planted forests, scattered crops, industrial and fruit trees, and industrial wood wastes. According to theoretical calculations, the total potential of biomass energy in Vietnam is more than 99 million tons/year, corresponding to an electric energy source of more than 340,000 GWh [17]. Biomass energy production in Vietnam is interested in developing alongside other forms of energy such as fossil fuels: coal, oil and gas, natural gas, solar power, and wind power. The expected biomass volume will increase rapidly in 2020, 2030, and 2050 by 1.8 MW, 9 MW, and 20 MW, respectively [18]. The use of biomass from agricultural by-products plays an important role in Vietnam in ensuring energy security, increasing the value of agricultural production, and tackling the environmental problems.

As Vietnam has favorable natural conditions such as hot and humid weather, lots of rain, and fertile soil, food crops proliferate. Therefore, the source of by-products from agriculture and forestry is abundant and continuously increasing. However, those by-products become the cause of environmental pollution such as burning forests, straw, sawdust, or dumping rice husks down rivers and canals. However, Vietnam has yet to exploit the potential of biomass energy production from agricultural by-products effectively. Biomass energy production still needs to be developed compared to its potential, and the commercialization of biomass energy production technologies is still limited. Moreover, biomass currently has been used on a small scale in rural areas and has unsuitable technology for production. In addition, policies to support the development of biomass have not been paid attention, followed by lacking financial and technical support for biomass production.

To bridge those gaps, this study conducts a qualitative analysis of the emerging economy of Vietnam to find out the motivations for biomass energy production. We obtain information from in-depth interviews with farmers, traders, and producers. On that basis, propose solutions for provinces in Vietnam that have renewable energy from biomass projects, replacing traditional energy sources to reduce CO₂ and protect the environment. The results of this study will contribute to renewable energy literature and support the formulation of policies in mainstreaming biomass energy production. The remainder of this paper is structured as follows. The literature review is presented in Section 2, which is followed by the methodology and data in Section 3. Results and findings from the bibliometric analysis are discussed in Section 4, while the implications are presented in Section 5. Section 6 concludes the paper with the limitation of the study.

2. Literature review

2.1. Renewable energy in agricultural production

According to the UN (2023), renewable energy is energy generated from natural sources or natural processes with a rate of formation higher than the rate of consumption. For example, sunlight and wind are sources of energy that are replenished daily. On the other hand, fossil fuels – coal, oil, and gas – are non-renewable resources and take hundreds of millions of years to form [19]. Burning fossil fuels to create

energy will release greenhouse gases that are harmful to the environment and people, such as CO₂ [20].

Renewable energy in agriculture also includes generating electricity to perform some farm tasks: pumping water for irrigation, for livestock, or for domestic use; lighting farm buildings; providing energy for processing and other operations ([21]; Zhng et al., 2023). These forms of renewable energy include solar energy, wind, and water power, oil from plants, wood from sustainable sources, other forms of biomass (plant materials), and biogas (gases) from the fermentation of manure and crop by-products) [22].

2.2. Biomass energy production from agricultural by-products

Following Brunori [23], Biomass can include products and by-products from agriculture and forestry. At the same time, Biomass can also include non-fossilized and biodegradable organic parts of industrial parks and municipal solid waste [24]. Biomass includes gases and liquids recovered from decomposing non-fossil organic materials (Demirbaş, 2001). Tursi [25] and Lu et al. [26] have classified agricultural Biomass into various types: Biomass from plants, animals, and microorganisms. Plant biomass is polysaccharides and vegetable oils, while animal biomass is polypeptides and animal oils, and Biomass from microorganisms comes from fungi and microalgae [27]. Otherwise, Biomass can also be classified according to the purpose of use, such as in direct burning to provide energy or converting into biofuels, including energy crops, sugar crops, starch crops, lignocellulose, vegetable oils, and animal fats [28].

Kozłowski et al. [29] expressed that energy plants include woody plants and grasses, in which woody trees are usually short-term forestry trees. Herbaceous plants include canola, sunflower, sorghum, and perennial grasses such as jasmine and thatch [29]. These are grown to produce energy through direct combustion or gasification to generate electricity and heat or convert them into liquid fuels [30]. Sugar crops include sugarcane, sugar beet, and sweet sorghum, while starch crops include corn, tapioca, wheat, potatoes, sweet potatoes, and rice [31]. These crops are grown to produce sugar or starch and use fermented ethanol as a biofuel. In addition, vegetable oils, animal fats, and oils from microorganisms are raw materials often converted into biodiesel (biodiesel) [32]. Tezer et al. [33] stated that biomass sources are converted into different forms of energy, such as electricity, heat, steam, and fuel for transportation by different conversion methods. Statuto et al. [34] also addressed that modern biomass energy technologies focus on using agricultural by-products as a fuel source for industry through combustion or gasification processes with low SO_x and NO_x. These sources can be by-products in the wood processing industry, agricultural by-products (bagasse, coconut shells, rice husks, rice straw, corn stalks, etc.), food processing industry (animal waste and by-products) with large output and acceptable calorific value [35].

Table 1 presents the chemical composition of several popular agricultural by-products. Depending on different biomass sources, the chemical composition can change, with the carbon component accounting for the highest proportion, followed by oxygen, hydrogen, and nitrogen [37]. Although the sulfur content is minimal, it will significantly affect biomass conversion [5]. The composition of carbon, oxygen, and hydrogen dramatically influences the energy of the generated biomass [6]. The higher the ratio of H/C and C/O, the higher the calorific value of the biomass [7].

This study focuses on biomass energy production from using agricultural by-products. An emerging economy, Vietnam, has the potential to use biomass electric energy because of abundant agricultural by-products such as rice straw, bran, rice husk, bagasse, cattle manure ... Statistics in 2016 of the Institute of Energy said that the capacity to collect agricultural by-products of the Mekong Delta is about 23 million tons/year, of which there are over 3.8 million tons of rice husks, nearly 17 million tons of rice straw, more than 372,000 tons of corn stalks; almost 1.4 million tons of waste sugarcane (Monre, 2016). The use of

Table 1
Chemical composition (%) and calorific value of biomass.

Biomass source	Carbon %	Hydrogen %	Oxygen %	Nitrogen %	Sulfur %	Tro %	Calorific value (MJ/kg)
Bagasse	44.8	5.3	39.6	0.38	0.01	9.8	17.33
Corn stalks	43.7	5.6	43.3	0.61	0.01	6.3	17.65
Straw	41.8	4.6	36.6	0.70	0.08	15.9	16.28
Barley Straw	45.7	6.1	38.3	0.40	0.10	6.0	17.25
Wheat Straw	43.2	5.0	94.4	0.61	0.11	11.4	17.51

Source [36]

biomass, especially agricultural by-products, plays a vital role in emerging economies to ensure energy security, increase the value of agricultural production, and solve environmental problems [38].

3. Methodology

3.1. Bibliometric analysis

Data for this study were collected from the papers published in the Web of Science database. The meta-data of published papers relating to biomass energy production from agricultural by-products including farmers' willingness toward biomass energy production was accessed. The retrieval time was on the 19th of March 2023. This study excluded publications from developed countries and only selected those from emerging economies. Emerging countries include Brazil, Russia, India, China, Colombia, Indonesia, Vietnam, Egypt, Turkey, South Africa, Chile, Czech Republic, Hungary, Malaysia, Mexico, Morocco, Philippines, Poland, South Korea, Taiwan, Bangladesh, Oman, Argentina, Greece, Israel, Qatar, Pakistan, Ukraine, and Thailand (Golden Gate University, 2023). This study only selected papers published in English and during the period from 1992 to 2022. As a result, the number of selected papers for the study is 314.

3.2. Qualitative analysis

This study conducted qualitative analysis by taking in-depth interviews with farmers, traders, and bioenergy producers. The selection of interviewees was based on their willingness to participate and their

accessibility. Five provinces in Viet Nam, which are Hoa Binh, Hanoi, Thanh Hoa, Hochiminh city, Daklak, and Gia Lai provinces.

For interviewing farmers, we randomly selected three communes in three provinces (Hanoi, Hoa Binh, and Thanh Hoa) to conduct interviews. In each commune, we obtained the list of farmers from the commune governments to select the participants. For interviewing traders, we employed the snowball method which started from a trader in Hochiminh city and then for four more traders from the contact of the first trader. For interviewing bioenergy producers, we also used the snowball method to kindly approach the manager of one bioenergy producer and ask for more contact with other three companies. The in-depth interviews were conducted with fifty respondents who are farmers, traders, and bioenergy producers.

4. Results

4.1. Results from bibliometric analysis

4.1.1. Analysis of citation

The published articles collected in our sample come from 29 emerging economies from 1992 to 2022. The number of publications is small from 1992 to 2014, dramatically increasing from 2015 to 2022 (Fig. 1). During this period, the number of publications in China was higher than in other countries (92 papers), followed by India (51) and Malaysia (29). The number of related publications in emerging economies like Vietnam, Qatar, Oman, Philippines, and Indonesia remains at the lowest rate.

The citation of documents on biomass energy production from

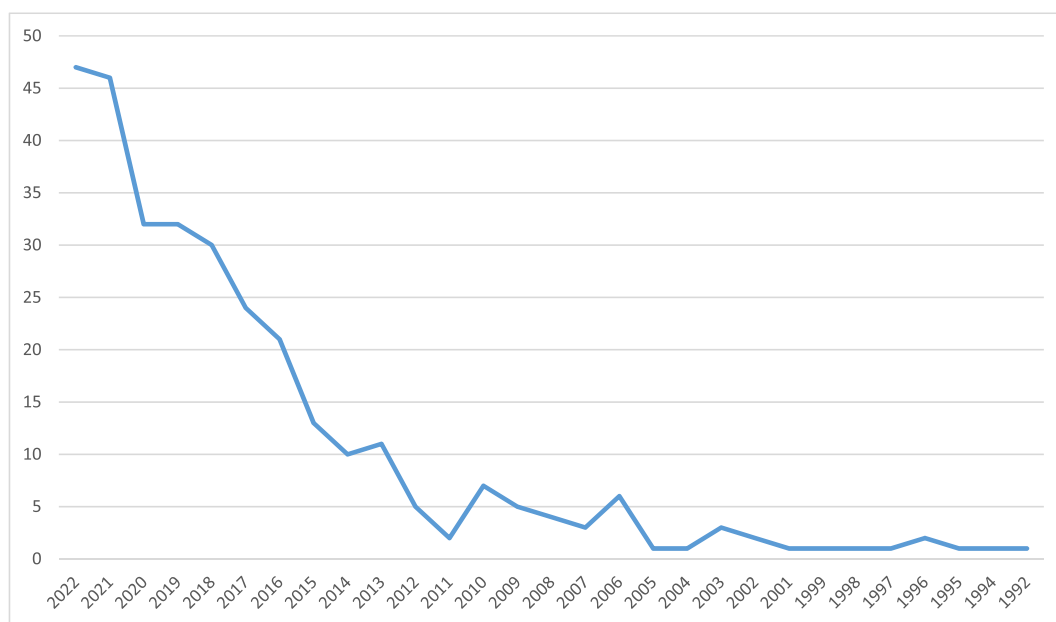


Fig. 1. The publications about biomass energy production from agricultural-by products from 1992 to 2022 (Authors' compilation)
Note: Vertical axis describes the number of publications. Horizontal axis describes the year of publications.

agricultural by-products was conducted. We set the minimum number of citations of a document to 1, making 259 papers meet the threshold. Ahmad et al. [39] have the highest citation, with 2730 citations in evaluating biomass and carbon stocks in the forests, which is followed by Snell and Peoples [40], with 354 citations in studying biomass-based bio-refineries (Table 2). Differently, Tsai et al. (2007) suggested the technology to produce bio-crudes from agricultural wastes.

4.1.2. Co-occurrence analysis

By analyzing the co-occurrence, this study identified keywords relating to biomass energy production in aquaculture. Table 3 shows that biomass is a more frequently used keyword, followed by pyrolysis, adsorption, bio-char, and activated carbon.

In density visualization, biomass has a high density and strong relationships with other keywords. Chen et al. (2016, 553) state that “the higher the density, the more mature and well-developed study on the theme”. This finding suggests that topics relating to biomass in agriculture are matured while other research fields are still immature (i.e., behaviors, willingness to pay, adoption, social acceptance, renewable energy) (Fig. 2).

The co-occurrence in Fig. 3 presents the keywords' network. This study finds that “biomass”, “waste”, “renewable energy”, “circular economy”, and “sustainability” have a close link. These keywords have matrix links with the others. In the overlay visualization, keywords in a different cluster have a different color. For example, the yellow shows that the keywords were frequently used in 2020. From 2014 to 2020, the previous studies focused on biomass, activated carbon, pretreatment, biofuels, rice husk. From 2020 to present, scholars have paid more attention to investigate circular economy, farmers' willingness to pay, digestion, and technologies.

These keywords are divided into nine clusters. Each cluster contains the same keywords and reflects the same topic (Chen et al., 2016). Each cluster represents a different aspect of managing biomass in agriculture (Table 4).

From the bibliometric results, this study comprehensively analyzed literature and data sources to identify the richest and most relevant literature sources on biomass energy production. Firstly, scholars have realized the perceptions of biomass's importance since 1992. The rapid expansion of single-species intensive aquaculture has caused severe environmental and socioeconomic problems (Folke & Kautsky, 1992; Tovar et al., 2000). Secondly, the benefits of biomass energy production from agricultural by-products have been well-defined since 2001. This finding is also demonstrated by Demirbas (2001), Ignaciuk and Dellink [49], and Zou et al. (2022). They all demonstrated the exploitation of agricultural by-products into biomass energy production to obtain stability in energy usage and economic benefits and to take full advantage of biomass resources and alleviate the energy crisis (Zou et al., 2022).

4.2. Results of qualitative analysis

The respondents' professional backgrounds range from farmers,

Table 2
The top ten of the highest citations from 1992.

Ranking	Document	Citations
1	Ahmad et al. [39]	2730
2	Snell and Peoples [40]	454
3	Tsai et al. [41]	368
4	Christoforetti et al. [42]	331
5	Girgis et al. [43]	314
6	Balat et al. [44]	267
7	Mench et al. [45]	260
8	Wang et al. [46]	203
9	Yaashikaa et al. [47]	180
10	Srinivas et al. [48]	176

Source: authors' compilation

Table 3
The frequently used keywords.

From 1992 to 2022	Occurrences
Biomass	69
Pyrolysis	29
Adsorption	25
Bio-char	21
Activated carbon	21
Removal	18
Waste	18
Lignocellulosic biomass	17
Pretreatment	16
Energy	16
Anaerobic-digestion	15
Fermentation	14
Gasification	13
Performance	14

Source: authors' compilation

traders, and bioenergy producers. Regarding the thirty-seven responses, farmer accounts for 75.6 %, producers accounts for 10.8 %, trader accounts for 13.6 %. The female respondents account for 58.5 % while male respondents account for 41.5 %. Most of the age of respondents are from 35 to 60.

4.2.1. Assessment about biomass energy production from agriculture by-products

Most farmers accept using biomass energy if the product quality is similar to or higher than the current energy quality and at an affordable price. In addition, this study also found that farmers are also willing to collect and provide by-products to traders for more income.

We currently treat rice husks/sugarcane stalks/bagasse by burning these wastes and get asked to fertilize trees and vegetables (Farm households in Hoa Binh).

We are willing to collect rice husks/sugarcane stalks/bagasse for traders/producers for more income (Farm households in Hanoi).

We don't want to buy this type of product because We rarely use electricity; we mainly use a straw for fuel (Farm households in Hoa Binh)

Therefore, this study highlights that the role of producers in manufacturing biomass energy at a reasonable price is a critical point. This finding is consistent with Amjith and Bavanish's [50] research results. They addressed that biomass and renewable energy need fair pricing mechanisms to be competitive with fossil energy.

Traders are willing to use biomass fuels/energy if the manufacturers provide quality products at a reasonable price. The quality of products is equivalent to that of fossil fuels on the market. If the quality at a lower price is not as good as fossil fuels, they will refuse.

We will accept to collect agriculture by-products from farmers and sell them to producers if their input buying process is stable and is committed (by-product traders).

It is very difficult to collect, transport and store biomass materials because agriculture by-products are not concentrated in a certain place. The cost of transporting and collecting raw materials are high (by-product traders).

We can import fuels from biomass energy production if producers make sure the biomass fuels are better than fossil fuels (by-product traders).

When traders accept the sale of this product, they will have many channels to advertise them to the market. However, they also fear the cost of collecting and transporting agriculture by-products. Ezz et al. [51] and Gani et al. [52] also addressed the importance of sustainable practices in biomass waste collection and processing. By exploring the

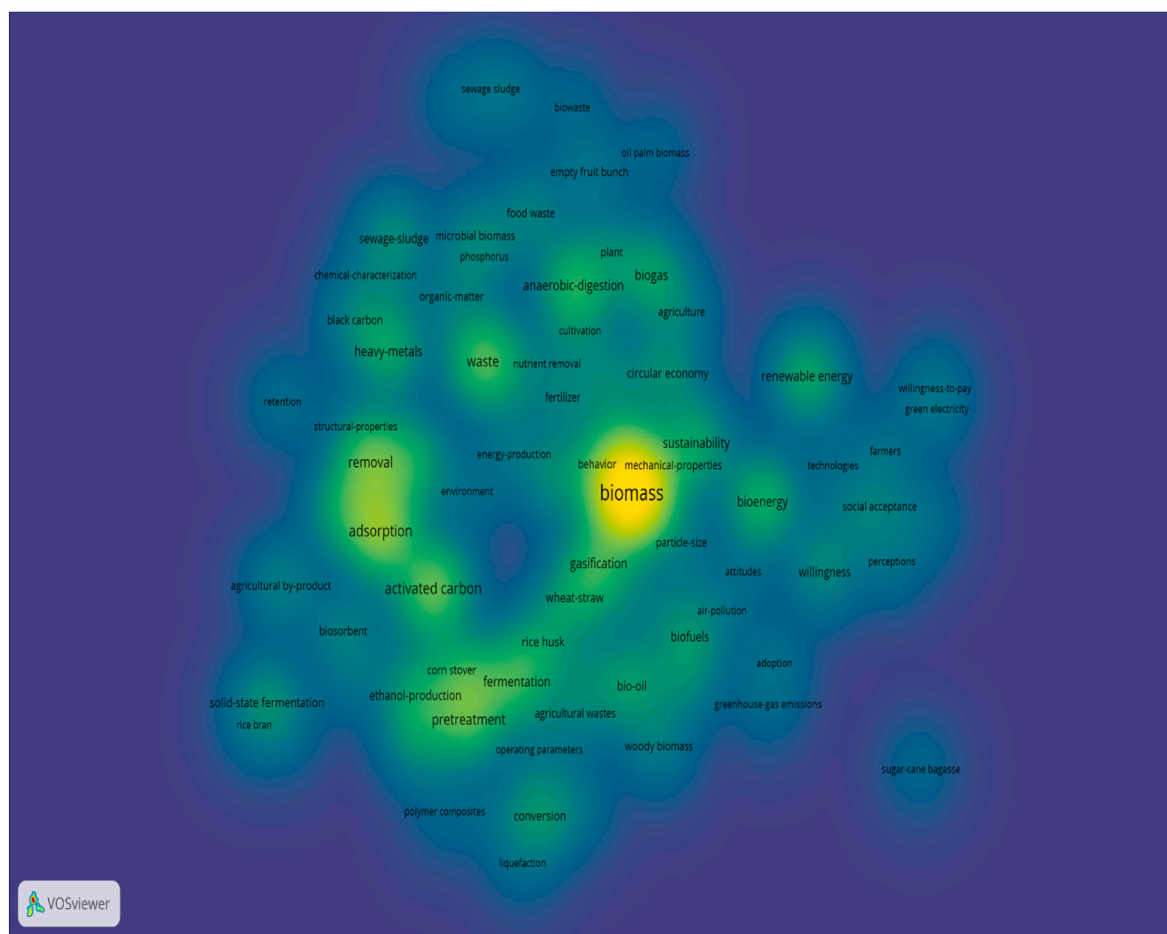


Fig. 2. The density visualization of keywords analysis.
Source: authors' compilation

abundant agricultural by-products into conversing biomass energy, emerging economies can limit fossil fuel dependence, reduce gas emissions, and improve sustainable development [53]. Producers are willing to make biomass energy from agricultural wastes if the users understand the benefits of such products and want to buy them.

We think producing biomass energy is feasible if we coordinate with farmers in collecting the input materials for production (Businesses 1)

We are facing the technology challenge. The high cost of biomass energy production technology is the biggest barrier. Furthermore, Vietnam has large coal reserves and low exploitation and transportation costs. At the same time, environmental taxes on thermal power are almost very low or non-existent. Therefore, biomass fuel is currently not competitive and cannot replace traditional fuel (Businesses 2).

We also have challenges in accessing the input (raw materials) because agriculture by-products are scattered in many provinces. Therefore, it is very difficult to collect, transport and store biomass materials to produce electricity. The cost of transporting and collecting raw materials has a significant impact to the cost of biomass electricity production. Consequently, suppliers of agricultural wastes are needed in our manufacturing process (Businesses 3).

In addition, producers are willing to cooperate with farmer households to ensure the supply of raw materials for production. This study aligns with Malladi and Sowlati's recognition (2018). They addressed that efficient collection and logistics systems and advanced biomass

energy production technology are critical to ensuring the economic viability and environmental sustainability of biomass energy production. Furthermore, the financial viability of biomass energy is examined by considering factors such as feedstock costs, energy efficiency, and market demand. Potential challenges and barriers, including technological, regulatory, and market-related aspects, are also discussed to provide a comprehensive overview of the feasibility of biomass energy production [54].

4.2.2. The micro conditions in producing biomass energy

Through our interview, we learned that most farmers know the harmful effects of environmental pollution and the importance of biomass energy. Moreover, knowledge of farmers about utilizing agricultural by-products has been uncovered through their confirmation of collecting agricultural wastes.

If we use biomass energy, we will protect environment (Farm households 1).

Using biomass energy will protect the environment and increase income for households who take advantage of by-products (Farm households 2).

Our study reveals that traders in an emerging economy know how to sell this biomass energy to contribute to protecting the environment. Our finding is similar to the study of Gani et al. [52]. They highlight that agricultural by-products (rice straw, corn stalks, palm oil) are addressed as the promising raw materials for biomass energy, which contribute to waste management, environmental concerns related to open burning and landfilling.

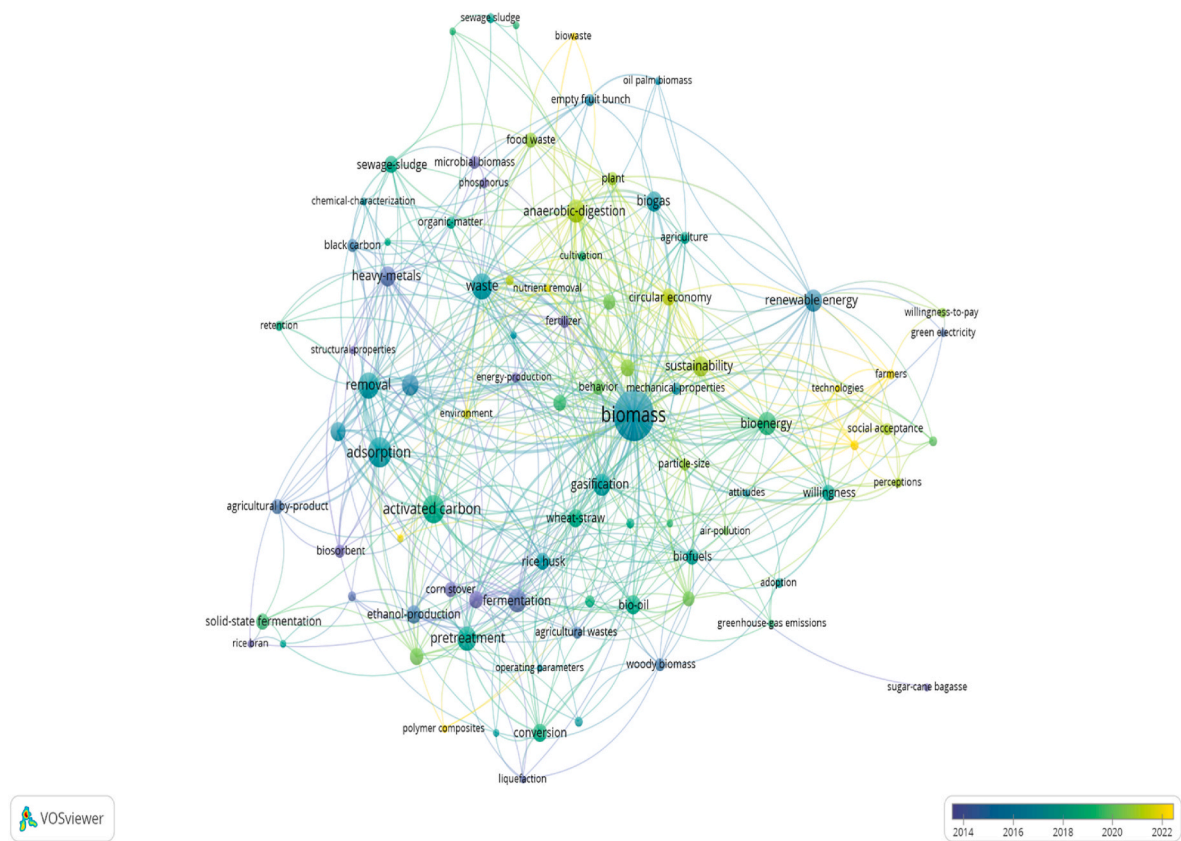


Fig. 3. The keywords’ network of publications about biomass energy.
Source: authors’ compilation

Clustering of keyword network.	
Cluster	Keyword network
1	Adoption, agricultural waste, attitudes, bio-oil, biofuels, conversion, gasification, greenhouse-gas emission, hydrogen-production, life-cycle assessment, sawdust, woody biomass
2	Anaerobic-digestion, bacteria, biogas, bio-waste, nutrient removal, sustainability, cultivation, energy-production, environment
3	Acid-hydrolysis, corn stover, ethanol-production, fermentation, operating parameters, pretreatment, rice straw, wheat bran
4	Bioenergy, energy transition, farmers, green electricity, perceptions, renewable energy, social acceptance, willingness to pay
5	Black carbon, empty fruit bunch, fertilizer, oil palm biomass, organic-matter, sewage-sludge, soil physical-properties, waste
6	Activated carbon, adsorption, agricultural by-product, bio-sorbent, chemical activation, heavy-metals, retention
7	Air pollution, behavior, biomass combustion, mechanical properties, microbial biomass, particle-size, phosphorus, rice husk, wheat straw
8	Agriculture, amended soil, biogas production, by-product, plant, sewage sludge, wet torrefaction
9	Biomass, sugar-cane bagasse

Source: authors’ compilation

We think that selling biomass energy is to protect environment if producers adheres to production techniques and raw material origins (Traders in HCM)

If farmers and many manufacturing companies have a full aware of biomass energy and want to buy, we think that this kind of business is potential for us (Trader in Daklak)

From producers’ interviews, we reveal that producing this kind of product will bring added value to businesses in enhancing the image of environmental protection. In other words, they all think that these

products contribute to protecting the environment.

We think that producing biomass energy is to protect environment (Businesses 1).

If we produce this product, we think we will have more reputation in raising people’s awareness about environmental protection (Businesses 2).

The research’s results found that factors motivating farmers to use biomass energy are quality and price.

We think we can transfer from using honeycomb charcoal to biomass fuels if this product has lower price than charcoal and is better than the current one (Farm households)

The logistics system and prices are motivations for traders in selling biomass energy. Regarding the prices of biomass energy, this result confirmed the findings of Deboni et al. [55], who mentioned how to calculate the final price. Differently, Pio et al. [56] highlighted the importance of showing the quality of biomass energy.

We think the motives for importing this kind of biomass energy are the quality and price in the value chain (By-product Trader 1).

We also think that the logistics system is important for all stakeholders in the value chain of biomass energy (By-product Trader 2).

Moreover, traders in buying biomass energy from producers and selling biomass energy to customers address the importance of sharing information, interactions, and relationships to increase the effectiveness of stakeholders in the biomass energy chain. This finding is in line with the results of Li et al. [57], which highlighted the role of the interactions among stakeholders in the value chain. Previous studies suggest that businesses will be willing to cooperate if the relationship is built on a foundation of trust and sharing (Liu et al., 2918 [58]).

We need the sharing of information from the bioenergy producers. This sharing is important for us to send the message to customers (By-product Trader 1)

To be successful in biomass distribution, I think interactions between parties in the supply chain are important to create sustainable social relationships (By-product Trader 2).

In a supply chain of biomass energy, businesses will be willing to cooperate and share resources to create added value when they believe that all members in the chain are taking action for common goals [59]. Birasnav et al. [60] also confirmed that social capital can create a trustworthy environment between supply chain partners and share information and resources, in which this sharing help stakeholders in the chain improve their creativity and innovation. Sharing information, building interactions and relationships among stakeholders in the value chain are considered as social capital [61]. Rahayu et al. [62] stated that social capital is widely accepted as an external resource that helps businesses gain competitive advantage.

We also found that producers' motivations in producing biomass energy are digitalization, raw materials and finished product prices, and government support. Regarding digitalization, previous scholars have addressed the critical role of digitalization because digital technologies have changed the firm operations [63].

To produce this product, we need to be assured of a stable supply quantity, quality, and price to produce biomass energy. We especially address that digital transformation is needed for our business (Businesses 1).

We think that digitalization in our business is also important for us to obtain the effectiveness and improve our performance (Businesses 1).

We need supports from government about credit, and the price of finished biomass energy (Businesses 2)

Other scholars have highlighted the importance of prices in evaluating the potential of biomass energy. Knápek et al. [64] mentioned three interconnected perspectives: (1) minimum price for growing biomass for energy purposes, (2) opportunity costs with respect to other uses of the arable land, namely production of conventional crops, and (3) the aspects of substitution of fossil fuels (e.g., coal) for biomass—the analysis of the demand. The main findings are that to give the farmers the same rate of return as conventional crops, the price of intentionally grown biomass for energy purposes in emerging economies would have to be higher than is the adequate minimum price. On the other hand, in order to be competitive as a substitute for coal for local space heating, the price of biomass would have to be close to zero [65].

4.2.3. The macro conditions in producing biomass energy

We found that government supports from farmers' perspective are related to credit support. Our findings are similar to the research of Zhang et al. [66]. They proposed the credit support from government for farmers in an emerging economy. Donner et al. [67] also argued that credit supports from government sets the boundary conditions for successful operations.

Government should support for farmers under tax reduction, low interest loans in agricultural production (Farm households)

Most traders believe that the government should reduce taxes, provide credit support, award certificates and organize seminars. This finding is in line with several studies in suggesting the government supports (Haque et al., 2021; Hasan et al., 2020; Hishamunda et al., 2009)

We think that the government should support the development of an effective logistics system to collect, store, ensure the properties of

biomass and transport it to conversion plants is a very important issue (Traders).

We need a credit support to import the manufacturing technology. We also need land incentives for project development, long-term power purchase contracts and environmental tax and other tax incentive policies (Businesses).

Most of bioenergy producers are well awareness of the importance of digital transformation. They desire the government support for them to implement digital transformation. Hrustek [68] also argued that digitalization improves the efficiency of biomass supply chains from precision cultivation to intelligent storage and logistics to optimal utilization by the end user. Since the energy sector started implementing innovative solutions and digital transformations, Borowski [69] analyzed production processes issues in organizations using solutions based on the paradigm of Industry 4.0; surveys were performed in the energy and food sectors.

Digitalization is important for us to increase the productivity and competitive advantages. However, we think that the government should create legal and infrastructure conditions for our businesses to easily carry out digital transformation (Businesses).

The government supports from our results are relating to credit and loans, the reduction in environment tax and other kinds of tax, the stable price of biomass energy, and upgrade infrastructure to encourage the digital transformation in business.

5. Implications and conclusion

5.1. Theoretical implications

This study reveals the motivating factors of biomass energy production from agricultural by-products: digitalization, technical and logistics systems, social capital, prices, and government support. These findings have contributed several implications to renewable energy literature. Renewable energy is being noticed in developing countries, but it still needs to be fully exploited [70]. Therefore, these research results serve as a lever and implications in further promoting renewable energy and biomass energy production from agricultural by-products.

Firstly, this study reveals that digitalization is important for businesses in producing biomass energy. This study also highlights the importance of technical and logistics systems in biomass production. This finding is in line with the suggestions of Gold and Seuring [71], Chidozie et al. [72], and Malladi and Sowlati [73]. They all demonstrated the critical role of logistics' contributions to both the material inputs and the costs of biomass energy production. In the emerging economy of Vietnam, agricultural production areas are widely distributed throughout the country. The types of biomass are not concentrated but distributed and scattered. Therefore, it is challenging to collect, transport, and store biomass materials for production. Once the logistics system has been controlled, biomass production will be developed.

Secondly, this study also implies that social capital and the prices of biomass energy are important for all stakeholders in the value chain. This finding supports the research of Miao and Khanna [8] and Wu et al. [74]. They addressed the fact that biomass prices are not currently competitive with traditional energy. This is particularly true for an emerging economy like Vietnam. The high cost of biomass energy production technology is the most significant barrier.

Moreover, Vietnam has large coal reserves and a low cost of mining and transportation. The environmental tax on thermal power needs to be higher or more present. Therefore, biomass fuel is not currently competitive and can replace traditional fuel. Furthermore, to develop biomass energy in the value chain, social capital between stakeholders in the biomass energy supply chain is considered an important asset that promotes accumulation, investment, and innovation in the supply chain and improves the efficiency of biomass energy production.

Thirdly, this study addresses the fact that the government plays an important role in supporting firms' access to credit for biomass production. In an emerging economy, one of the current challenges is the need for more capital to produce biomass energy [75]. Many investors, banks, and developers consider renewable energy projects in Vietnam as needing help to access credit [76]. And the important role of policies and institutions have been considered in developing bioeconomy [12–16]. Hence, biomass energy production projects need a guarantee from the Government and project sponsor.

Finally, this study found the role of government in supporting firms in the logistics system, the price to produce biomass energy, and facilitating businesses in digital transformation. These findings support other studies about the importance of government support, such as North and Pienaar [77] and He et al. [9]. The distance between the biomass feedstock and the plant site determines the transportation cost and risk of delay or loss of the feedstock. This factor has proven to have an impact on the cost-benefit ratio of the project. While using storage facilities can address part of the risk of this factor, it creates an additional cost for biomass energy. Consequently, the government needs to develop an efficient logistics system to collect, store, and preserve the properties of the biomass and transport it to the conversion plants, which is very important in the development of biomass production.

5.2. Practical implications

This study contributes several practical implications for stakeholders (farmer households, traders, and bioenergy producers) in the value chain of biomass energy production. This study also suggests some implications for government and policy-makers in developing biomass and renewable energy.

Regarding business implications, bioenergy producers need to improve awareness about digital transformation at all levels of the organization, from non-managerial employees to top managers, to implement a digital transformation. Firms also develop training programs on digital transformation for staff. For technical production, firms should choose biomass energy production technology suitable for raw materials, usage goals, and the region's specificity, which is crucial for optimizing and improving energy efficiency. Besides, firms need to optimize logistics and enhance the selection of materials used. In detail, firms need to optimize the transportation and storage system, help ensure a stable supply of energy materials, minimize the deposition of fixed ash in the boiler, and avoid production interruptions. The quality of the raw materials used in biomass production is essential. Good quality raw materials will help ensure the efficiency of the boiler's operation, minimizing the possibility of irritating the boiler. Therefore, it is necessary to use the best possible materials. Finally, through this research, stakeholders in the biomass energy value chain can gain an overall view of the importance of social capital. Biomass energy producers can increase interaction with partners, strengthen trust, share awareness and a culture of cooperation with supply chain partners to exchange information and resources for innovation. Firms can see that social capital plays a vital role in promoting innovation and improving business performance in supply chains. Therefore, social capital will become an essential strategy for companies. The results of this study also recommend that policymakers should use social capital to develop risk management strategies and operational strategies.

Regarding policy implications, governments in emerging economies like Vietnam should focus on building tools and guidance documents, implementing awareness-raising activities, and providing systematic digital transformation training for biomass energy producers. Moreover, the government needs to promote reform and improve the connectivity capacity of stakeholders in the biomass energy value chain. To ensure the sustainable prices of biomass energy, the government should raise the electricity price for all biomass energy technology so that the industry can reach its expected potential capacity. Secondly, the government encourages project developers to consider new funding methods,

such as Special Purpose Companies, to attract more investment and study the mechanism of attracting social investment in the transfer system to develop energy research projects. Domestic banks need to improve their capacity to handle renewable energy projects. Thirdly, the government needs specific and long-term strategies, mechanisms, and policies to support the mobilization of investment capital from state, private, and international budget sources for research, deployment, and biomass energy development. To enhance renewable energy development, the government must have attractive and favorable policies and mechanisms for investors, such as land incentives for project development, long-term power purchase contracts, and environmental tax policies and policies. Finally, from the study's findings in a certain developing country (Vietnam), to develop the technical capability, knowledge transfer, and financial supports, governments in developing countries need to seek the international collaboration. Having cross-country collaboration, developing countries gain the knowledge sharing, the international financial supports, and best practices and lessons from energy policies from other countries, especially high-income countries, in resolving common problems in the pursuit of sustainable energy goals as the suggestion of Falcone [14].

5.3. Limitations

Biomass energy production has great potential to replace fossil energy sources and ensure energy security. It is widely drawing attention of scholars, businesses, and governments. Digitalization, technical and logistics, social capital, prices, and government supports are the main determinants to develop biomass energy in Vietnam. To strengthen these revealed factors, it is important to developing countries having international collaboration in knowledge sharing, international financial supports, international training, and best lessons from other countries. This study also has implications for developing countries in general in developing biomass energy.

However, this study still has several limitations. Firstly, the bibliometric analysis of this study uses data collected from 1996 to 2022, excluding early cited articles and languages other than English. Future research should include articles that are early cited and non-English papers for better results. Secondly, the data was gathered from the Web of Science database. Future research may export data from other journal databases such as Scopus. Finally, this study conducted in-depth interviews with farmers, traders, and bioenergy producers. There was a lack of consumers and enterprises using biomass energy in manufacturing. Future studies should also study the role of enterprises and consumers to have broader insights into biomass energy production.

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CRediT authorship contribution statement

Thi Khanh Chi Nguyen: Conceptualization, Methodology, Writing – original draft, preparation. **Nguyen Quoc Toan:** Validation, Reviewing and Editing.

Declaration of competing interest

We wish to draw the attention to the Editor to the following facts that there is no conflict of interest exists. We also confirm that We have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

Data availability

Data will be made available on request.

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